NAME OF STUDENT_____

MZUZU DIOCESE

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

MATHEMATICS

PAPER II

(100 marks) Subject Number: 131/II

DATE: 27th March Time allowed:7:30-10:00am (2hours 30 mins)

Instructions

1. **This paper contains 19 pages. Please check.**
2. Answer all **Six** questions in **section A** and any four questions from **section B**.
3. **Section A** carries **60 marks** and **section B** carries **40 marks**
4. The maximum number of marks for each answer is indicated against each question.
5. Write your answers in the spaces provided on the question paper
6. Calculators may be used.
7. **All working must be clearly shown.**
8. Write your name at the top of each page on your question paper in the spaces provided.
9. In the table provided on this page, **tick** against the question number you have answered.

Question number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
Total			

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Section A: (60 marks)
Answer all the seven questions in this section

1. a. Simplify the expression $\frac{x^2-y^2}{x^2+ab+ax+by} \div \frac{x^2-2xy+y^2}{cx+bc}$

(4 marks)

b. Without using a calculator, simplify $\frac{2\sqrt{8}}{\sqrt{96}-2\sqrt{6}}$, giving the answer with a rational denominator. **(4 marks)**

2. a. The second term of an arithmetic progression is three times the seventh term, and the ninth term is 1.find the first term and the common difference. **(6 marks)**

b. The probability a day will rain heavily is $^{5}/_{8}$, the probability of flooding when it rains heavily is $^{6}/_{7}$ and the probability of flooding when it rains lightly is $^{1}/_{5}$. Use the tree diagram to find the probability of flooding. **(5 marks)**

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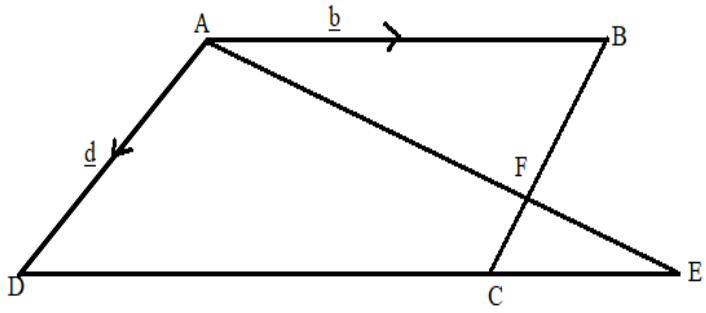
3. **a**. In the figure below, ABCD is a parallelogram. $\overrightarrow{DC} = 4\overrightarrow{CE}$, $\overrightarrow{AB} = \underline{b}$ and $\overrightarrow{AD} = \underline{d}$.



Show that $\overrightarrow{AE} = \frac{1}{4}\left(5\underline{b} + 4\underline{d}\right)$

(5 marks)

b. Prove the theorem which start that the angle between the chord and a tangent at the point of contact is equal the angle subtended by the same chord in the alternate segment. **(5 marks)**

4. **a**. Solve the equation $12x^3 + 16x^2 - 5x - 3 = 0$. **(5 marks)**

b. The length of the triangle is 8 cm and 15 cm, if the included angle is $60°$. Find the area of the triangle and leave the answer in a surd form. **(5 marks)**

5.**a**. **Table 1** shows the weight of some students in a certain school.

Mass (Kg)	34.5	44.5	54.5	64.5	74.5	84.5
Number of students	4	26	40	26	2	2

Calculate the standard deviation. **(6 marks)**

b. Solve the equation $log_2(x^2 + 3x - 2) = 2log_2 x + 1$. **(4 marks)**

6 **a**. The sum of the squares of the ages of the mother and the daughter is 964 years. Four years ago, the difference of their ages was 22 years. Find their present ages. **(6 marks)**

b. Using a ruler and a pair of compasses only, construct in the same diagram,

i a circle centre O of radius 4 cm. **(1 marks)**

ii a point **R** outside the circle such that **OR**$= 10\ cm$. **(2 marks)**

iii Tangents **TR** and **XR** to the circle at **T** and **X** measure and state the length of the tangents. **(2 marks)**

SECTION B (40 marks)

Answer any four questions from this section

7. An aeroplane leaves airport A on a bearing of $N24°E$ and flies for 340 km to another airport B. it then leaves airport B and flies on a bearing $N64°W$ to another airport C. If the airport A and airport C are 680 km apart, calculate the bearing of airport A from airport C.

(10 marks)

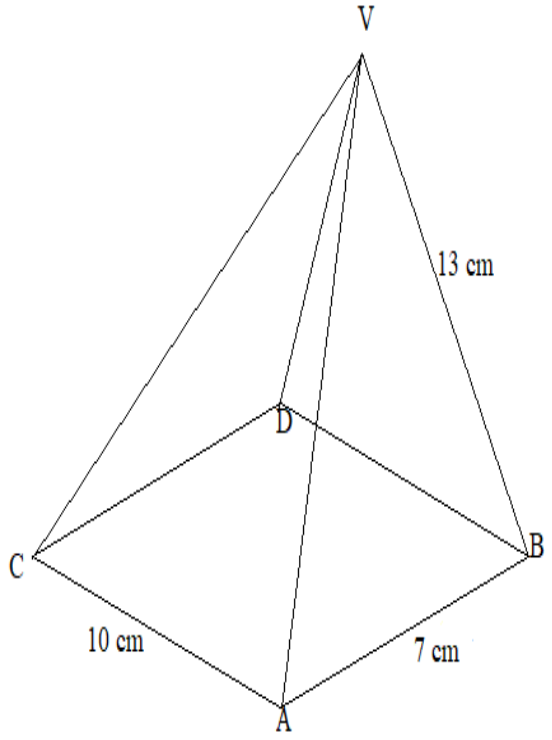
8. The distance (d) covered by a moving object is partly constant and partly varies with the square of its velocity (v). When the velocity is 2m/h, the distance is 15m and when the velocity is 3m/h, the distance covered is 20m. If the distance covered is 36m, find the velocity of the object. **(10 marks)**

9.There 60 bags of fertilizer which contains at least one of the following nutrients: Nitrogen (N), Potassium (K) and Phosphorous (P).

- 12 bags contain Nitrogen only.
- 7 bags contain Potassium only.
- 11 bags contain Phosphorous only.
- 6 bags contain all the three nutrients.
- $n(K \cap P \cap N') = x$
- $n(N \cap P \cap K') = 2x$
- $n(N \cap K \cap P') = 3x$

Using a Venn diagram, calculate the number of bags which contain Nitrogen. **(10 marks)**

10.The figure below shows a right rectangular based pyramid such that $VB = AV = CV = DV = 13\ cm, AC = 10\ cm$ and $AB = 7\ cm$



Calculate the angle between the face VAB and the base **(10 marks)**

11. A certain boarding school had K5, 000,000.00 to buy bags of rice and bags of Maize flour. A bag of rice costs K50, 000.00 and a bag of flour cost K20, 000.00. The school decided to spend money as follows:

- Buy at least 30 bags of rice and at least 50 bags of flour
- Spend more on rice than on flour by not more than K1, 000,000.00

i. Taking x to represent the number of bags of rice and y to represent the number of bags of maize flour. Write down four inequalities in x and y that satisfy the above information. **(4 marks)**

ii. Using a scale of 2cm to represent 20 units on the horizontal axis and 2 cm to represents 25 units on the vertical axis, draw a graph to show the region bounded by the four inequalities by shading the unwanted region. **(5 marks)**

iii. Use your graph to find the maximum number of bags of maize flour the school can buy if 40 bags of rice have already been bought. **(1 mark)**

12. **The table 2** shows some of the values for y= $x^2 - 3x + 10$

i. Complete the table below. **(2 marks)**

Table 2 y= $x^2 - 3x + 10$

x	-1	0	1	2	3	4	5
y	14	10	8		10		20

i. Use the scale of 2 cm represent 1 unit on the x-axis and 2 cm to represent 5 units on the y-axis, draw the graph of y= $x^2 - 3x + 10$ **(5 marks)**

ii. Use the graph solve equation simultaneous equation **(3 marks)**

$y = x^2 - 3x + 10$

$y = x + 7$

END OF QUESTION PAPER